

defects. If there are features in the acoustic image caused by echoes from unexpected interfaces, the part may have an anomaly.

Fig. 1 shows four of the anomaly types that may be found when a single boardmounted BGA package is scanned. All four types are gaps that contain air. Each has a solid-to-air interface that reflects nearly all the arriving ultrasound.

The white oval at the upper left is an isolated void in the mould compound. It is a fairly large bubble of air, but its potential for causing damage is likely small because it is not near wires or other key items. It is unlikely to expand in size. It is discussed separately below.

The white feature in the die attach material is a risk because it will block heat radiating from the die and could cause the die to overheat and fail. It might also expand because it is close

to two solid surfaces having different thermal expansion characteristics.

The third feature is not a bubble-like void but a delamination between the mould compound and the die. Delaminations can be only a few microns thick; the thickness of the delamination shown here was exaggerated to make it visible.

The fourth feature at the right end of the die is a crack. Thermal and mechanical stresses make it very likely that an internal crack like this will damage the circuitry above.

The area surrounding the void in Fig. 1 is shown in more detail in Fig. 2. The scanning transducer that pulses ultrasound into the part is out of sight above the part. The leftmost arrow represents a pulse arriving from the transducer and striking the mould compound-to-air interface at the centre of the void. Because the interface at this x-y location is horizontal, virtually all of the pulse is sent back to the transducer as a high-amplitude echo.

The second pulse strikes an angled interface, and much of it is scattered in directions away from the transducer. The percentage of the pulse reflected to the transducer depends on the location of the point of reflection on the downward sloping interface. Some of the echoes

will reach the transducer but will be imaged in a colour such as blue.

The third pulse misses the void and strikes the interface between the mould compound and the die. The interface between these two solids reflects about 60% of the pulse to the transducer. The remaining 40% of the pulse travels into the die and will be reflected from the next material interface it encounters.

What the diagram does not show is the amount of data that is collected by scanning. If the void is 1mm wide, the number of x-y locations from which echoes are received during a single pass across the void approaches 100, but may be more if higher resolution is required. The speed with which ultrasound travels through production materials is typically very high. As the transducer is scanning, the first pulse to encounter the void will send back an echo and the echo will be received.

The next x-y location in line will not be pulsed until after an interval. The transducer will collect echoes from all x-y locations where there are material interfaces.

In the BGA package in Fig. 1, the interfaces at the top and bottom of the die are key defect sites. The same is also largely true of other package types, such as the two QFPs shown in the following figures.

Fig. 3 shows one region of a printed circuit board, which was first imaged optically in black and white. The two largest components were then scanned individually by a C-SAM acoustic micro imaging tool, and the acoustic images were then superimposed on their optical images in this figure. Colour in the acoustic images is determined by the amplitude and polarity of the return echoes.

As long as it can be accessed by the acoustic tool's transducer, any flat item on a printed circuit board can be imaged. For example, the numerous large and small ceramic chip capacitors in Fig. 3 could be imaged to find internal cracks or voids. The AMI tool is programmed to recognise each board type, and can navigate to accommodate components of varying thicknesses.

Fig. 4 is the acoustic image of the lower left IC package in Fig. 2. The items of interest are the two red fea-

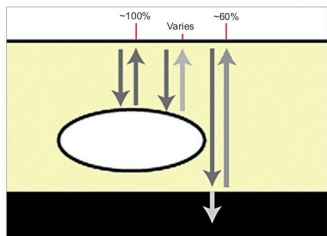


Fig. 2: Enlarged diagram of void

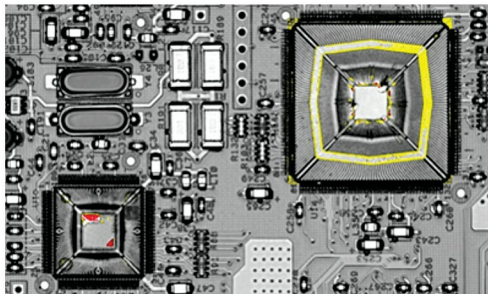


Fig. 3: Acoustic images (which use colour) of two QFPs superimposed on a monochrome optical image